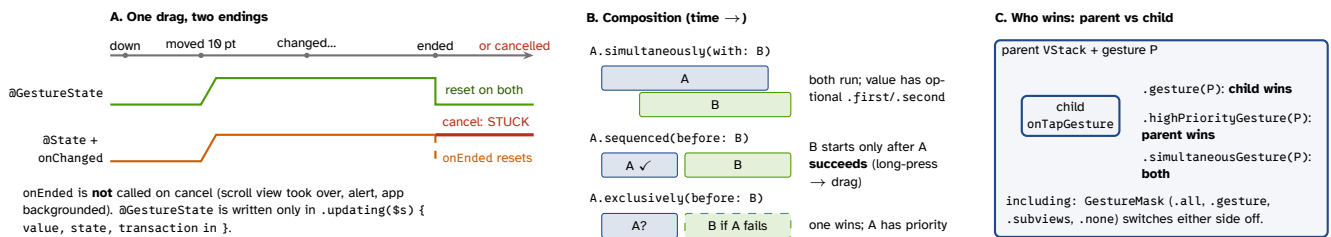


In one line: Gestures are **value types** composed with combinators (replacing recognizer-delegate arbitration); **@GestureState resets on end and cancel**. **Shape** = one vector view, **Canvas** = one immediate-mode view for thousands of primitives, **drawingGroup()** = flatten a subtree into one offscreen image before display.



Gestures

- **Types** (...Gesture): Tap(count:), SpatialTap (16, has location), LongPress(minimumDuration:), Drag(minimumDistance:) (default 10 pt), Magnify / Rotate (17; were Magnification / Rotation).
- **Callbacks**: .updating (→ @GestureState, auto-reset), .onChanged (→ @State, you reset), .onEnded (commit; success only). Animated snap-back: @GestureState(resetTransaction:).
- **Drag values**: translation is a delta from startLocation; location is absolute in the chosen space (.local, .global, .named(...)).
- **Hit testing**: transparent areas (Spacer, Color.clear) don't receive touches → .contentShape(Rectangle()); .allowsHitTesting(false) lets touches pass through.
- **ScrollView vs drag**: scroll pan and your drag compete — a plain .gesture(DragGesture()) blocks scrolling or is swallowed. Fixes: bigger minimumDistance; .simultaneousGesture; long-press .sequenced(before:) drag (the reorder pattern); .scrollDisabled(isDragging) (16); UIGestureRecognizerRepresentable (18) for delegate arbitration / direction checks.

Example — drag that snaps back on cancel

```
@GestureState private var drag: CGSize = .zero // transient
@State private var pos: CGSize = .zero // committed
RoundedRectangle(cornerRadius: 12)
    .offset(x: pos.width + drag.width, y: pos.height + drag.height)
    .gesture(DragGesture(minimumDistance: 20)
        .updating($drag) { v, state, _ in state = v.translation }
        .onEnded { v in // success only
            pos.width += v.translation.width
            pos.height += v.translation.height })
```

Drawing

- **Shape**: path(in rect: CGRect) returns a Path; rect from layout. Path: move(to:), addLine, addQuadCurve, addCurve, addArc, closeSubpath. Without animatableData (AnimatablePair for two) it snaps, not morphs. stroke straddles the path (half is clipped at the edge); strokeBorder (InsettableShape) stays inside.
- **Canvas { ctx, size in }** (15): immediate mode — no view per element, no diffing. GraphicsContext is a **value**: copy it to scope a transform/opacity; fill, stroke, draw, clip(to:), addFilter, blendMode, drawLayer.

ctx.resolve(Text(...)); views passed as symbols: + .tag(id) → resolveSymbol(id:).

- **TimelineView(schedule)** (15) re-runs its content on .animation(minimumInterval:paused:), .periodic(from:by:), .everyMinute, .explicit; read context.date. The system throttles it when not visible — cap the rate for battery.
- **drawingGroup()**: composites the subtree into one offscreen bitmap. Wins for many overlapping / blended / shadowed views; costs an offscreen texture (memory), is rasterised (blurs if scaled), and Core Animation-composited views inside (controls, web/media, most UIKit) show a placeholder.

Canvas vs views vs drawingGroup

Views	tens of elements; each tappable, accessible, transitions
drawingGroup	must stay views but compositing is the cost
Canvas	hundreds-thousands of visual-only primitives: charts, particles, waveforms

```
TimelineView(.animation(minimumInterval: 1/30)) { tl in
    Canvas { ctx, size in // ONE view, 5 000 dots
        let t = tl.date.timeIntervalSinceReferenceDate
        for p in points { ctx.fill(Path(ellipseIn: CGRect(
            x: p.x, y: p.y + 4 * sin(t + p.x), width: 3, height: 3)),
            with: .color(.green)) }
        if let s = ctx.resolveSymbol(id: 0) { ctx.draw(s, at: .zero) }
    } symbols: { Image(systemName: "star.fill").tag(0) }
}
```

Traps

- ! @State in onChanged without handling cancel — the view stays mid-drag.
- ! Tap + long-press on one view fight — state the relation with .exclusively(before:).
- ! Mixing translation (delta) with location (absolute) or the wrong coordinate space.
- ! **Canvas is opaque**: no per-element hit testing, no VoiceOver — add .accessibilityLabel / .accessibilityChildren { }, do hit tests yourself from the gesture location.
- ! drawingGroup() everywhere — memory + blur for nothing; measure first.

Remember

“GestureState forgets, State remembers; Canvas draws, views are things.”